ENRICO LAOH, Ph.D.

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Brief About Me

- I am an **interdisciplinary researcher with a PhD in Industrial Engineering and Management**, working at the **intersection of artificial intelligence, information technology, life science analytics,** and **managerial aspects,** with a focus on developing **adaptive, interpretable, and human-centered intelligent systems to support decision-making in complex systems**. My research advances **AI-driven decision frameworks** by integrating **machine learning, explainable AI, and optimization** to enable reliable, transparent, and adaptive decision-making.
- My **current** research spans **healthcare and biomedical analytics, genomics analytics, environmental analytics, and intelligent decision support systems**, emphasizing the translation of complex multimodal data into **actionable insights for sustainable and evidence-based decisions**.
- Supported by a strong record of **peer-reviewed publications, international awards, competitive fellowships, and leadership experience**, including guiding an **INFORMS student chapter to national recognition**, I am committed to advancing **interdisciplinary and translational AI research** with real-world impact and commercialization potential, reinforced by recognition as a **NASA-funded I-Corps Creativity, Innovation, and Entrepreneurship Scholar**.

Educations

Doctor of Philosophy, OKLAHOMA STATE UNIVERSITY — July 2026
Focus Area/Major: Industrial Engineering and Management
Degree GPA: 4.00/4.00

Graduate Certificate, OKLAHOMA STATE UNIVERSITY — July 2023
Focus Area/Major: Business Analytics and Data Science
Degree GPA: 4.00/4.00

Master of Science, OKLAHOMA STATE UNIVERSITY — December 2022
Focus Area/Major: Industrial Engineering and Management
Degree GPA: 4.00/4.00

Master of Engineering, UNIVERSITAS INDONESIA — July 2017
Focus Area/Major: Data Engineering and Quality
Degree GPA: 4.00/4.00

Bachelor of Engineering, UNIVERSITAS INDONESIA — July 2016
Focus Area/ Major: Industrial Engineering
Degree GPA: 3.88/4.00

Certifications: *Tableau Desktop Specialist*, *Amazon SageMaker Studio*, *Amazon Redshift*, *AWS MLOps*, *Six Sigma Green Belt*

Selected Awards and Honors

- **Graduate Student Leadership Award**, School of Industrial Engineering and Management, 2025
  Recognizes a graduate student who has demonstrated outstanding leadership abilities in student organizations.
- **Lisa Zaken Award for Excellence**, Institute of Industrial and Systems Engineers, 2024
  This international annual award, given only to one winner who has outstanding scholastic achievement, leadership, and dedication to the Industrial and Systems Engineering profession, demonstrates the highest of standards.
- **Third place in the 2024 IISE ESD/QCRE/PG&E Energy Analytics Challenge**, 2024
  Annual international competition by the Energy Analytics Society, judged by academia and industry practitioners.
- **President Leadership Recognition**, Hargis Leadership Institute, 2024
  Awarded for demonstrating exemplary leadership and impact within the university and broader community.
- **Outstanding Graduate Student Award**, CEAT, Oklahoma State University, 2024
  Selected as the top graduate student in the CEAT based on academic excellence and research achievements.
- **Robberson Dissertation Fellowship**, Oklahoma State University, 2024
  Recognized for outstanding doctoral research contributing to advancements in industrial engineering and data science.
- **Roy and Virginia Dorrough Distinguished Graduate Fellowship**, Oklahoma State University, 2023, 2024, 2025, 2026
  Awarded to recognize outstanding graduate students with evidence of exceptional academic performance and achievements, leadership experiences, and notable extracurricular or community engagement activity.
- **Creativity, Innovation, and Entrepreneurship Scholar**, Spears School of Business, Oklahoma State University, 2021
  Distinguished student with excellence, innovativeness, creativity, responsibility for change, engagement in one's surroundings, personal integrity, and strong moral character.
- **IEEE SMC Society Merit Paper Awards**, 8th International Conference on Awareness Science and Technology (iCAST), 2017
  Given to the most outstanding paper in the IEEE SMC academic society annually that is presented at the conference.
- **IEEE Indonesia Section Best Presenter Award**, 3rd International Conference on Science in Information Technology (ICSITech), 2017
  Given to the best-presented paper at the conference.

Work Experiences

COMPUTATIONAL GENOMIC RESEARCH SCIENTIST — January 2026 to **present**

TRANSLATIONAL GENOMICS LABORATORY, OKLAHOMA STATE UNIVERSITY, USA

Collaborate with multidisciplinary omics teams to integrate multi-omics data and study the pathway from DNA to RNA to proteins and phenotypes. Apply computational modeling to convert large-scale biological data into testable hypotheses, leveraging expanding genomic resources. Advance scalable, whole-genome, and spatially adaptive approaches to improve phenotype assessment and variety development.

RESEARCH ASSOCIATE — January 2026 to **present**

FACULTY OF AGRICULTURAL LIFE & ENVIRONMENTAL SCIENCES, UNIVERSITY OF ALBERTA, CANADA

Analyze large-scale genomic and phenotypic data to investigate genetic variation,

population structure, and trait associations in forest tree species, particularly lodgepole pine and white spruce. Apply Ancestral Recombination Graphs (ARGs), GWAS, gene annotation, and evolutionary analyses to identify genetic markers and biological mechanisms associated with forest growth and resilience. Develop a genomically informed Multi-Agent Reinforcement Learning (MARL) framework to support adaptive forest management, integrating genetic insights, ecological sustainability, and economic objectives.

DATA SCIENCE EXPERT — August 2019 to **present**

INDUSTRIAL ENGINEERING VENTURE UNIT, UNIVERSITAS INDONESIA, INDONESIA

Provide optimal consulting and training services by leveraging expertise for private/government institutions in Indonesia.

DATA SCIENCE INTERN — June 2024 to August 2024

UNITED AIRLINES INC., CHICAGO, USA

Observed the pricing main model and understood the effect of demand forecasting. Updated the demand forecasting model and tested novel variables extracted from internal databases and external potential influences. Reduced forecasting *MAPE by 2%.*

HEALTH DATA ANALYTICS RESEARCHER — October 2022 to July 2026

CENTER FOR HEALTH SYSTEMS INNOVATION, OKLAHOMA STATE UNIVERSITY, USA

Developed frameworks and models to analyze healthcare data and synthesize knowledge to support a medical-based decision-making system. *Funded by a $1.2M NIH Grant*.

TEACHING AND RESEARCH ASSISTANT — January 2021 To December 2025

SCHOOL OF INDUSTRIAL ENGINEERING AND MANAGEMENT, OKLAHOMA STATE UNIVERSITY, USA

Taught tutorials and courses in Industrial Engineering, Operations Management, and Data Analytics. Did research on collecting, extracting, and building interpretable and explainable Machine Learning models from human physiological data (EEG, ECG, Eye Motion, etc.) to measure stress levels and cognitive state to help industrial AI-based systems generate robust decisions.

NON-TENURE TEACHING PROFESSOR and RESEARCHER — August 2017 to December 2020

INDUSTRIAL ENGINEERING DEPARTMENT, UNIVERSITAS INDONESIA, INDONESIA

Researched the application of descriptive, predictive, and prescriptive analytics to a wide range of industrial problems, including network optimization, power generation, customer segmentation, natural language processing, and supply chain design. Taught and conducted classes related to industrial engineering and data analytics.

Professional Memberships

- **Institute for Operations Research and the Management Sciences (INFORMS)**
  OSU Chapter President (2022-2023): Won the cum laude chapter award. Member since 2019.
- **Institute of Industrial and Systems Engineers (IISE)**
  Member since 2021.
- **American Medical Informatics Association (AMIA)**
  Member since 2025.
- **Eastern North American Region International Biometric Society (ENAR IBS)**
  Member since 2026.